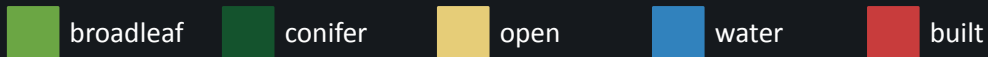


VISTA RETREAT · 15 SEPTEMBER 2026

Scandibérique Map Challenge

Ride 3 km along the Canal du Loing. Take photos. Rebuild a bird's-eye map of the whole valley from 30 of them and your wits.



row 0 = north



col 100 = east

How the day goes

1

afternoon

Ride

Each team photographs the route for itself. Location on. Take plenty, you pick 30 later.

2

at the break

Check

We read the EXIF on a few phones per team: is the GPS in the files?

3

evening · 2.5 h

Build

Own photos, no maps. Choose your 30, turn them plus reasoning into a map.

4

deadline

Submit

One map file and the 30 photos per team. Run the checker first.

5

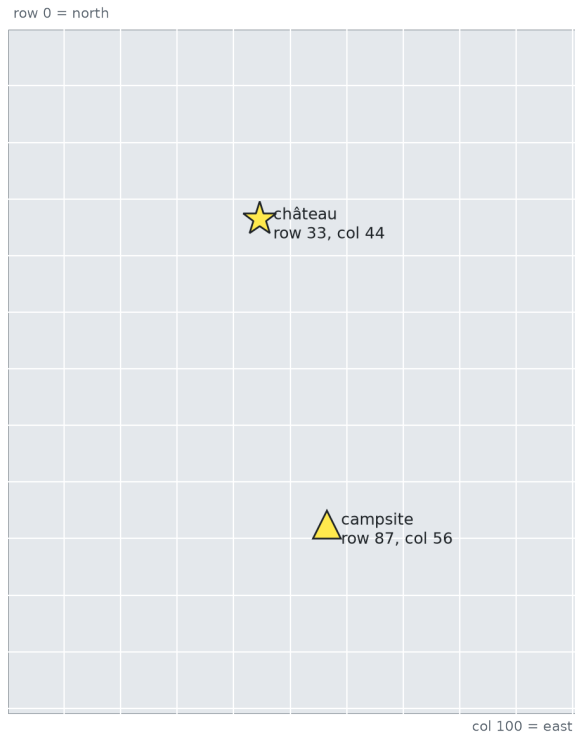
after

Reveal

Scores, leaderboard, then the ground truth over satellite imagery. We argue.

Two things are measured, and both count: what you saw, and what you inferred about the rest of the valley.

The map is a grid



- **121 rows × 101 columns**

50 m cells. About 5 × 6 km around the ride.

- **Row 0 is north, column 0 is west**

Nemours in the north, the campsite in the south-east.

- **Lambert-93 (EPSG:2154), metres**

meta.json has the origin and the exact formula; kit.py has the function.

- **One class per cell, 1 to 5**

Unknown (0) counts as wrong. Fill everything, guess if you must.

- **The ground truth is hidden**

OpenStreetMap, rasterized the same way. You get it after scoring.

Five classes, one per cell

1 broadleaf Wide-leaved woods: oak, beech, hornbeam, birch, chestnut. A closed leafy canopy, trunks that branch.	2 conifer Needle-leaved woods, here almost all Scots pine. Dark, straight bare trunks, often planted in rows, a floor of needles.	3 open No tree cover, not water, not built: crops, stubble, meadow, orchard, cemetery, pitch, park lawn, scrub.	4 water The Loing, the Canal du Loing, lakes, gravel-pit ponds. <i>Not a stream you can step over.</i>	5 built Whole residential areas, houses with their gardens, industry, retail, car parks, rail land. <i>A road under ~8 m wide is not built by itself.</i>
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Priority: water › built › forest › open, for whatever covers a quarter of the cell; inside forest, the commoner leaf type wins.

No “mixed” class: a genuinely mixed stand takes the leaf type of the nearest stand that is not one.

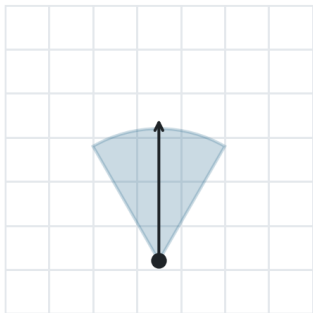
From the saddle: wide leaves = broadleaf. Straight dark trunks and needles = conifer. Fields = open. River = water. Houses = built.

Things you will actually meet

what you see from the bike	class	why
oak and beech, a leafy canopy overhead	broadleaf	the default wood in this valley
straight bare trunks, needles underfoot	conifer	Scots pine, usually planted, often in rows
a few pines among the oaks	broadleaf	no mixed class: whichever leaf type covers more
the towpath through the woods	broadleaf	a path under 8 m does not make a cell built
the canal beside you, the Loing across it	water	each is about one cell wide
the campsite	built	itches, buildings, car park: a built area
a village street with houses and gardens	built	the whole residential area, gardens included
a country road between two fields	open	the fields win; the road is too thin
stubble, maize, ploughed earth, pasture	open	all open, whatever grows there
an orchard, a cemetery, a park lawn	open	no continuous tree cover
a flooded gravel pit or a pond	water	if it is bigger than a cell
a hedgerow between fields	open	too thin to count as forest

What you see, and what you infer

scoring convention with heading:
60° cone, 150 m



scoring convention, no heading:
50 m disk

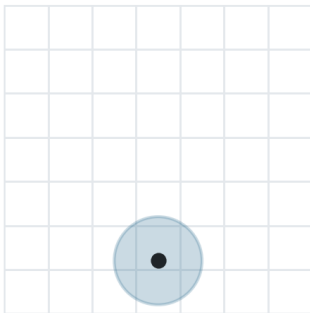
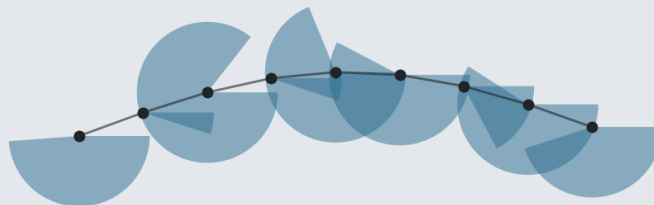


Photo budget: 30 per team

- Take as many as you like; hand in 30. Those define what you saw.
- How far to trust a photo is your call. The cone on the left is only our scoring convention, not physics.
- Prior scores use the cells neither team saw. Where you point the camera is part of the game.

observed: what you saw



unobserved: what you inferred (most of the map)

Most of the map is unobserved. Rivers follow valleys, towns sit on rivers, forests have shapes. That prior is the interesting half.

Three sub-problems worth separating

1 Recognise

What land cover is in each photo?

Zero-shot CLIP, a small VLM, colour statistics, or the LLM reading captions. Broadleaf against conifer is the sharpest call a photo makes, and the easiest to get wrong at a distance.

`photo → one of five classes`

2 Project

Where on the grid does it belong?

A photo has a position and maybe a heading. What it shows is in front of the camera, at some depth, on one side of the path. The river in the frame continues for kilometres. Foreground and background land in different cells.

`label + pose → cells`

3 Densify

From 30 cones to 12 000 cells

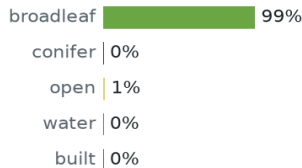
Thirty photos cover a few percent of the map. Rivers are lines, towns are blobs around a centre, fields and forest come in patches. Propagate, smooth, and let the LLM's knowledge of the Loing valley fill the rest.

`sparse cells + prior → dense map`

The unobserved score measures the third one. The observed score mostly measures the first two.

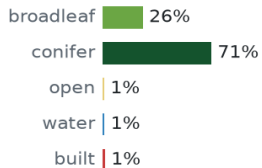
What a photo actually tells you

cyclamen under broadleaf trees



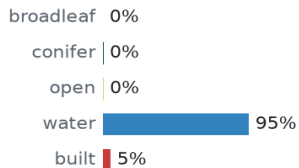
correct

a pine trunk against a boulder



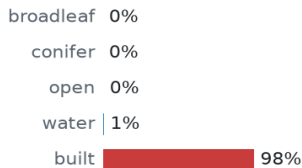
the leaf type, called right

the canal



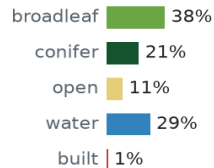
correct

a street — and the canal



one label per photo

a mossy rock



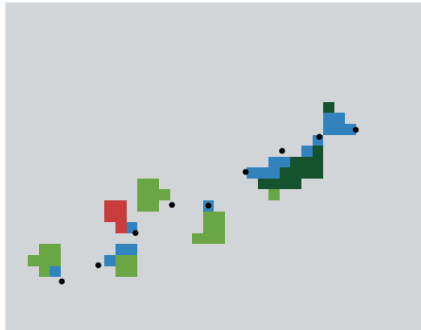
nothing to go on

Zero-shot CLIP on five photos from our own test run. No training, a few seconds on a laptop CPU.

Those 36 photos came out 21 built, 8 water, 5 forest, 2 open — and both “open” were photographs of paintings. Nobody points a camera at a field. Yours will be starved of open ground unless you aim at it on purpose.

From 30 cones to 12 000 cells

1 · what the photos gave



9 view cones, 5 % of the cells

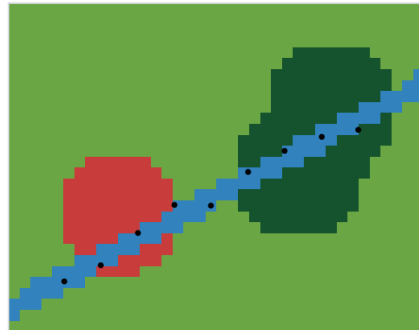
2 · nearest observed cell



every cell copies its closest observation

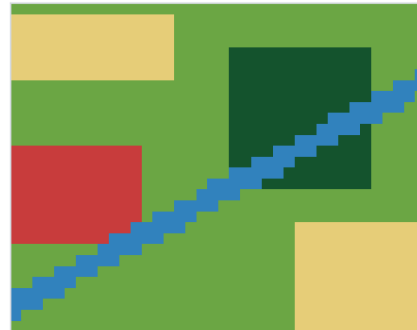
Toy region invented for this figure — not the real one. Panel 3 beats panel 2 everywhere the photos never reached.

3 · shapes from a prior



river = a line; town, fields, pines = blobs

the answer



hidden from you until the end

Nearest neighbour is the floor

Every cell copies its closest observation. Cheap, and it smears each photo across a whole wedge of the map.

Shapes do better

A river is a line you can extend. A town is a blob around a centre. Fields and forest come in patches. Fit those, do not copy cells.

Ask the guide

The LLM knows what a Loing-valley village looks like from above, which side the forest is on, how wide a canal is. Turn its answers into shapes.

Rules

Allowed

- Any program you write: CLIP or any vision model, colour heuristics, geometry, rules. Humans write code, prompts and parameters.
- Any LLM, online or offline, with web search and browsing switched off. Its memory is your prior: ask it what the Loing valley looks like.
- Geometry, EXIF headings, timestamps, your memory of the ride.

Not allowed

- Painting cells by hand. The map must come out of your program, fed by the photos and the LLM's answers.
- Anything that shows or describes a map: Apple Maps, Google Maps, OpenStreetMap, satellite imagery, Wikipedia, search engines.
- An LLM with search or browsing on. That is a map with extra steps.
- Photos not taken on today's ride, or more than 30 of them.
- The ground truth. You get it after scoring.

It is a game. Nobody checks your phone. The point is that the map comes out of a pipeline you built, fed by the photos and your head.

The kit, and what you hand in

git clone — four files

`README.md`

the grid, the classes, the rules, the format, ten lines to a first map

`meta.json`

the grid contract: shape, origin, class ids, palette

`kit.py`

EXIF photos → grid cells; paint_cone, fill_rest, nearest_fill, save, and the observed mask

`slides.pdf`

these slides

You hand in, per team

1 `<team>_submission.npy`

A 121×101 uint8 array, values 1–5. `save()` in `kit.py` writes it and prints the histogram.

2 `<team>_photos/`

At most 30 originals with EXIF, taken today. We compute your observed cells from these.

Every cell filled with 1–5. Unknown (0) is scored as wrong, so guess rather than leave it blank.

```
pip install numpy pillow pyproj scipy · python kit.py check teamA_submission.npy
```

Scoring

metric	on which cells	what it tells
mean IoU, 5 classes	all cells	the map ranking
mean IoU, 5 classes	cells neither team saw	the prior ranking
mean IoU, 5 classes	your observed cells	how good your perception was
the same, leaf types merged	all cells	what broadleaf vs conifer cost you
pixel accuracy	all cells	the obvious number

Two rankings: map (all cells) and prior (the cells neither team saw). IoU is averaged over the five classes, so water counts as much as broadleaf even though there is far less of it.

0.07

mean IoU of painting the commonest class everywhere. The floor. Beat it.

“Observed” is a convention

From your 30 photos we count as seen a 60° cone 150 m deep along the EXIF heading, or a 50 m disk with no heading. Same rule for both teams; it only decides which cells go in which split.

Before you ride

Stop the bike before you take a photo.

No shooting one-handed while rolling, and pull off the path before you stop. It is a towpath with other people on it, and nothing in this game is worth a crash.

1 Thirty photos per team, not per person

The cap is for the whole team. Agree on the bike who covers what: both sides of the path, and a few pointed away from it across the fields and the water. You pick the final 30 in the evening, so shoot more than you need.

2 Your photos already carry their own GPS

Keep location on for the camera. That EXIF tag is the only thing that tells us where a photo belongs on the grid, and a photo without it is dead weight. Send the originals: chat apps and “optimised” copies strip it out.

Roughly

afternoon	ride and photograph
at the break	we check the GPS is in your files
evening	2.5 h, two teams, build the map
deadline	map + your 30 photos
after	scores, the answer, argument